DEVINE NYAENYA

Cybersecurity Engineer · Security Builder Kenya
GitHub: github.com/Nyaenya-Devine **Portfolio**: devine-nyaenya-portfolio.vercel.app

Build → Test → Break → Learn → Secure

PROFILE

Self-taught security builder focused on application security, access control, and security engineering. Designs and ships defensive controls that are implemented and tested — not just described — then breaks them in authorized labs and hardens them. Open to security roles.

CORE SKILLS

SECURITY

- ■ Access control & RBAC (least privilege, separation of duties)
- ■ Tamper-evident audit logs (hash chains, HMAC)
- ■ Application security & secure software development
- ■ Threat modeling (assets, attackers, controls, detection)
- ■ AI-agent security — OWASP ASI03 (identity & privilege abuse)
- ■ Vulnerability assessment in authorized labs
- ■ Network security & Linux privilege escalation (labs)
- ■ SOC/defensive concepts (detection, alerting, SIEM)

ENGINEERING

- ■ Python — security tooling & labs (stdlib-first)
- ■ TypeScript · Next.js/React (App Router, full-stack)
- ■ SQL (SQLite/PostgreSQL + Drizzle) · REST API design
- ■ Testing: Vitest, pytest — controls proven by tests
- ■ Git & GitHub — commits, PRs, CI workflows
- ■ HTML/CSS/Tailwind — accessible, security-minded UI
- ■ Crypto primitives applied: PBKDF2, Argon2id, TOTP, HMAC-SHA256
- ■ Web security: CSRF, rate limiting, CSP, secure sessions

SELECTED PROJECTS & LAB WORK

◆ **Chokepoint — access-control & tamper-evident audit platform** ***[Live demo · flagship]***
github.com/Nyaenya-Devine/Nyaenya-Devine-chokepoint · live: nyaenya-devine-chokepoint.vercel.app

- ■ Least-privilege access control for high-impact operations by humans and AI agents: RBAC policy gate, dual-control two-person approvals, SHA-256 hash-chained + HMAC-signed audit ledger, explainable anomaly detection (after-hours, unknown sources, privilege escalation, automation).
- ■ 26 automated tests prove the security properties; built on Next.js 16 + TypeScript + Vitest; framed around the OWASP Agentic AI Top 10 (ASI03 — Identity & Privilege Abuse).

◆ **Android Reset Lab — simulation of an enterprise MDM reset system** ***[Simulation / lab]***
github.com/Nyaenya-Devine/android-reset-lab

- ■ Python simulation (never touches real devices) that prevents single-person abuse: default-deny RBAC, four-eyes dual-control, TOTP MFA, CSRF + rate limiting, 15-min lockout, hash-chained + HMAC audit log with SIEM shipping.
- ■ 52 tests; attacked my own system with 6 techniques — 6/6 detected (9 precise alerts); mapped to MITRE ATT&CK (T1110/T1078/T1134/T1070) and NIST 800-53; P0–P3 hardening fixed 15 bugs incl. timing attacks and enumeration.

◆ **Android Device Management Tool — real-device path (experimental WIP)** ***[Experimental / WIP — honest status]***
github.com/Nyaenya-Devine/android-device-management-tool

- ■ Evolving the reset-lab into a dual-mode Android Enterprise console (Next.js + PostgreSQL + Drizzle): local simulator plus live Google Android Management API integration — OAuth2 service-account JWT, encrypted keys at rest, CloudDPC QR enrollment, policies, WIPE/LOCK/REBOOT commands with correct acknowledge-before-wipe semantics, deprovision via enterprises.devices.delete.
- ■ Web + Python CI; go-live runbook written; live path implemented but not yet proven on real hardware — documented rather than claimed.

◆ **Security lab — authorized-environment write-ups** ***[Learning, documented honestly]***
portfolio → /security-lab

- ■ Hands-on exploitation write-ups in isolated, authorized cyber-range VMs only: full chain recon → foothold → privilege escalation → defensive lessons (e.g., UnrealIRCd backdoor CVE exercise).

Also built: EndoPima Kenya — bilingual (EN/SW) community-first endometriosis early-recognition prototype (privacy-conscious, local-first); built to learn health-tech product thinking beyond security.

CERTIFICATIONS & TRAINING

✓ **Ethical Hacker** — Cisco Networking Academy
✓ **Introduction to Cybersecurity** — Cisco Networking Academy
✓ **Python Essentials 1** — Cisco Networking Academy

Further training: Python Essentials 2 (Cisco curriculum) · Python Programming · Coding with AI · Create AI Agents with Copilot Studio (Microsoft) · AI for Beginners

Self-taught security engineering path; learning is verified by shipped code, tests, and lab write-ups rather than certificates alone.